Cross-Domain Few-Shot Learning Based on Decoupled Knowledge Distillation for Hyperspectral Image Classification

Shou Feng, *Member, IEEE*, Hongzhe Zhang, Bobo Xi, *Member, IEEE*, Chunhui Zhao, Yunsong Li, *Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— Existing cross-domain few-shot learning (FSL) methods for hyperspectral image (HSI) classification have garnered widespread attention due to their excellent performance in recognizing novel classes. To mitigate domain shift, researchers focus on designing sophisticated domain adaptation (DA) modules to directly apply biased metaknowledge in the target domain (TD). However, this paradigm proves somewhat inadequate in the face of significant differences in distribution. To cope with this dilemma, we adopted a new mindset of treating metaknowledge extraction and debiasing from the source domain (SD) as a synergistic process and proposed a cross-domain FSL framework based on decoupled knowledge distillation for HSI classification (HSIC). In general, to efficiently acquire and utilize unbiased metaknowledge, this framework centralizes on a knowledge distillation (KD) strategy. Through the effective information transfer process, the extraction and debiasing of metaknowledge were integrated into a comprehensive and productive process. Simultaneously, to release the constraints imposed by the coupled logits in the KD process on the knowledge interaction, the decoupled logit interaction (DLI) module is employed in the framework. This module decouples the traditional KD into two controllable components, making a more balanced and comprehensive interaction of task-related knowledge and data-intrinsic knowledge between models. Moreover, to facilitate the extraction of critical discriminative metaknowledge from the abundant redundant information in HSI, the discriminative information refinement (DIR) module is designed to develop distinctive features for similar bands. Extensive experiments on three public HSI datasets exhibited the superior performance of the proposed cross-domain few-shot learning method based on decoupled knowledge distillation for HSIC (DKD-FSL) method in comparison with seven state-of-the-art approaches.**

***Index Terms*— Cross-domain, few-shot learning (FSL), hyperspectral image classification (HSIC), knowledge distillation (KD).**

I. Introduction

THE hyperspectral image (HSI) is a 3-D data cube with abundant spatial and spectral information, containing unique characteristics of various land covers [1], [2], [3]. This superiority has led to widespread attention and applications in military and civilian fields, such as military target detection, precision agriculture, and mineral exploration [4], [5], [6].

HSI classification (HSIC) is one of the fundamental and crucial tasks in HSI analysis [7], [8], [9]. In early research, traditional methods primarily relied on extracting handcrafted features or employing classifiers for classification [10], [11], [12], [13], [14]. However, these methods depend heavily on the prior knowledge of researchers and exhibit significant limitations in representing high-dimensional data.

During recent years, deep learning (DL) achieved spectacular success in the remote sensing community [15], [16], especially in HSIC. This is mainly because these methods demonstrate an outstanding ability to capture deep and nonlinear features. Following the stacked autoencoder (SAE) was first applied for HSIC, various outstanding network architectures have been introduced to capture deep spectral features within HSIs [17], [18]. These architectures include convolutional neural network (CNN) [19], graph convolutional network (GCN) [20], [21], transformer network [22], [23], and others. Compared to traditional methods, DL approaches have obvious advantages in terms of feature representation capability, handling complex data, and generalization ability. However, the promising performance is closely related to the number of annotated samples. Owing to the prohibitive costs of the annotation process, reliable and sufficient annotated samples are invaluable for HSIs. Therefore, relaxing the demand for mass-labeled data in high-precision classification

Received 8 July 2024; revised 19 September 2024; accepted 3 October 2024. Date of publication 8 October 2024; date of current version 18 October 2024. This work was supported in part by the Open Fund of State Key Laboratory of Integrated Services Networks under Grant ISN25-05; in part by the National Natural Science Foundation of China under Grant 62471155 and Grant 62371153; in part by China Postdoctoral Science Foundation under Grant 2023M740265; and in part by the Fundamental Research Funds for the Central Universities under Grant 3072024XX0801, Grant 3072024XX0805, and Grant 3072024LJ0804. *(Corresponding author: Bobo Xi.)*

Shou Feng is with the College of Information and Communication Engineering and the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China, and also with the State Key Laboratory of Integrated Service Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: fengshou@hrbeu.edu.cn).

Hongzhe Zhang and Chunhui Zhao are with the College of Information and Communication Engineering and the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China (e-mail: zhanghongzhe@hrbeu.edu.cn; zhaochunhui@hrbeu.edu.cn).

Bobo Xi and Yunsong Li are with the State Key Laboratory of Integrated Service Networks, School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: xibobo@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Jocelyn Chanussot is with the GIPSA-Laboratory, CNRS, Grenoble INP, Université Grenoble Alpes, 38000 Grenoble, France, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: jocelyn@hi.is).

Digital Object Identifier 10.1109/TGRS.2024.3476116

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

has become one of the principal challenges facing the HSIC task.

Faced with such a dilemma, researchers have attempted some intuitive thoughts, such as expanding the training samples and employing low-parameter models [24], [25], [26], [27], [28]. While these two approaches alleviate the limitations of DL methods to some extent, they often fail to reconcile model performance with generalization capabilities. Meanwhile, a variety of new learning strategies based on semisupervised learning and self-supervised learning (SSL) have been proposed. These methods require in-depth mining of prior information in HSIs, whether from annotated or unannotated samples [29], [30], [31].

The success of SSL and semisupervised learning strategies has prompted scholars with new inspiration: whether the prior knowledge can be leveraged in other completely different HSIs. Such a situation where training and testing data come from two different domains is commonly referred to as cross-domain HSIC, and these two domains are source domain (SD) and target domain (TD). However, additional prior knowledge from the SD is not readily available. The primary obstacle is the inherent discrepancy in distribution that exists between the two domains. This discrepancy is determined by various complex factors involved in the data acquisition process. Even if two images are captured by the same sensor, differences in acquisition angle, time, geographical factors, and weather can cause spectral shifts within the same land cover type. To mitigate the performance degradation caused by the discrepancy, researchers have attempted to realign the distributions of the two domains at the feature level [32], [33]. By employing domain adaptation (DA) methods, the shared knowledge acquired from the SD can be utilized for the classification of the same classes in the TD [34], [35]. Nevertheless, due to the assumption that the two domains possess the same label space, DA methods remain ineffective when encountering novel classes in the TD.

In recent years, classification with scarcity and novel samples has garnered widespread attention due to its promising application prospects. Few-shot learning (FSL) has gradually evolved into an independent and novel research direction [36], [37]. Currently, the majority of FSL methods employ an episodic learning strategy introduced by metalearning. In particular, a training process can be divided into two stages: metatraining and metatesting. During the metatraining stage, contrary to traditional DL models, each episode focuses on diverse metatasks that are independent and diverse from each other. This "learn to learn" process guides the model toward acquiring the metaknowledge of distinguishing between different categories rather than specific class distinctions, which endows the model with strong generalization capabilities [38], [39]. However, because of the fact that metatraining and metatesting data typically originate from two distinct HSIs, this distributional shift still poses a seriously detrimental effect on FSL-based methods.

In the face of this critical situation, Li et al. [40] first conceptualized the transfer relationship between base and novel classes as a domain shift in transfer learning, introducing a novel task called cross-domain FSL (CDFSL). They adopted an adversarial-based DA module to address this challenge. Zhang et al. [41] attempted to mitigate domain shift by achieving alignment at both the feature and distribution levels through the graphs. Nevertheless, it is necessary to note that regular DA strategies are predicated on the assumption of shared feature space and label space between the source and TDs, which is not well-suited for handling data with salient distribution differences. This is particularly problematic in the presence of novel classes, as it can lead to inaccurate feature-level alignment. In this context, due to the fact that a huge amount of SD data in the metatraining is absolutely dominant, the distribution aligned through the DA strategy will inevitably lean toward the SD. Consequently, the model's predictions on the TD will be naturally biased toward the SD, leading to misclassifications [42]. From the perspective of FSL, the DA modules enable the biased metaknowledge to be applied in the TD by aligning the distributions of the SD and TD. However, these methods are not efficient and direct. The key to addressing CDFSL should lie in how to acquire unbiased metaknowledge, also known as discriminative knowledge, in a more reasonable and straightforward manner.

Beyond that, another focus of CDFSL research has been placed on how to exploit metaknowledge through more effective means. The straightforward approach taken by researchers is to establish effective representations of extensive raw data with better models or training strategies [43], [44], [45]. However, these methods do not take into account the data characteristics of HSIs and tend to underestimate the interference caused by the substantial redundant information. In the absence of an understanding of the relationship between bands, it remains uncertain whether models can actually unearth the real gems from the massive redundancy. Therefore, such an approach of solely enhancing model representation capabilities can hardly be deemed efficient in CDFSL for HSIC [46], [47], [48]. In fact, the crucial discriminative knowledge for classification is hidden among the many bands of HSI. If the model could be guided to acquire this kind of auxiliary information, it would be more beneficial to the task.

To address the above issues, we propose a more natural mindset for CDFSL: the extraction and debiasing of metaknowledge from SD should be a holistic and collaborative process. Following this line of thought, a cross-domain few-shot learning method based on decoupled knowledge distillation for HSIC (DKD-FSL) is proposed for HSIC. Specifically, the DKD-FSL method adopts the knowledge distillation (KD) framework as a cornerstone. As a high-performance technique for knowledge extraction and transfer, KD can enhance the model's discriminative ability while debiasing the model's observations of TD data from SD. Subsequently, we further divide the metaknowledge into task-related knowledge related to classification decisions and data-intrinsic knowledge linked to data properties. In order to avoid incomplete knowledge transfer caused by the coupling of task-related knowledge and data-intrinsic knowledge in conventional logit distillation, the decoupled logit interaction (DLI) module is introduced in this framework. On the basis of preserving sensitivity to task-related knowledge, the student model can enhance its focus on the data-intrinsic knowledge of the TD. Furthermore, to assist the model in extracting crucial knowledge from the extensive redundancy of HSI in a targeted

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

way, the discriminative information refinement (DIR) module is proposed to enhance the distinctiveness of discriminative information.

The main contributions of the article can be summarized as follows.

1) A CDFSL framework based on decoupled KD is proposed for HSIC. This framework unifies the extraction and debiasing of metaknowledge into a synergistic process, aiming to achieve efficient utilization of unbiased metaknowledge in predictions of TD. With KD as its core strategy, the framework guides the student model to focus on crucial characteristics of the TD data and gradually reduces the biased metaknowledge.
2) To avoid the deficient interaction by conventional logit distillation, a DLI module is introduced in the proposed framework. Depending on whether it is directly related to the probability of the target class, this module decouples the conventional logit distillation into two weight-controlled parts, which represent the interaction of task-related knowledge and data-intrinsic knowledge between models, respectively. The balance between these two components will establish a comprehensive knowledge interaction scheme within the framework.
3) To reduce the difficulty of acquiring discriminative metaknowledge from redundant bands in HSI, we proposed a DRI module. This module minimizes the distance between similar bands in a feature graph-based manner, which is fruitful in providing complementary and emphasized representations for similar bands.

The remainder of the article is structured as follows. The proposed DKD-FSL method is elaborated in Section II. In Section III, experimental results and relevant parameters of the proposed method are comprehensively analyzed. Finally, Section IV consists mainly of a summary of the article.

II. Methodology

In this section, the overall flowchart of the proposed DKD-FSL method is first briefly presented, which consists of three stages, as shown in Fig. 1. Subsequently, three stages are specifically described in order. In addition, the significant modules in each stage are introduced separately, including the DIR module and the DLI module.

A. Overall Framework

The overall flowchart of the proposed DKD-FSL is depicted in Fig. 1. Initially, the metatask construction and data augmentation for the SD and TD data are performed during the data preparation stage. In the metaknowledge extraction phase, all data are first processed by the DIR module, which standardizes the channel dimensions while enhancing the discriminative information representation. Subsequently, the FSL strategy will be employed on the student model to mine shared metaknowledge. At the same time, the student model, the teacher model, and the historical student model will predict different logits for the same TD data. In the KD stage, the predictions of the teacher model and historical student model will serve as regularization for the student model, enabling iterative elimination of bias from SD. The DLI module will decouple conventional logit distillation into two components to provide balanced adjustment for task-related knowledge and data-intrinsic knowledge in the knowledge interaction.

B. Data Preparation

To obtain transferable metaknowledge through the FSL strategy, two HSIs are involved in the metalearning training process: the SD $\mathcal{D}^s$ and the TD $\mathcal{D}^t$. In the construction of metatasks, n^s samples $\mathbf{X}^s \in \mathbb{R}^{p\times p\times B^s}$ will be randomly selected from $\mathcal{D}^s$ with C^s different categories, while n^t samples $\mathbf{X}^t \in \mathbb{R}^{p\times p\times B^t}$ are sourced from $\mathcal{D}^t$ with C^t classes, where p is the size of patches and B^s and B^t represent the number of bands in SD and TD. Taking the SD as an example, N samples will be selected from C classes in each episode, where $C = C^t$. These samples will be partitioned into support sets $\mathbf{X}_s^s = \{\mathbf{x}_i^s, y_i^s\}_i^{C\times N_s}$ and query sets $\mathbf{X}_q^s = \{\mathbf{x}_j^s, y_j^s\}_j^{C\times N_q}$, where N_s and N_q represent the number of support sets and query sets.

In traditional KD, controlling temperature is commonly used to adjust soft labels [49], [50]. However, the process of determining the optimal temperature is often cumbersome [51], [52]. To simplify the process, the proposed method allows the student model to learn from strongly augmented images, while the teacher model learns from weakly augmented images. By artificially creating a difference between the predictions of the two models, this approach increases the diversity of information received by the student model while guiding it to focus on the inherent properties of the images. Assuming that there are N^t labled samples in TD, for the raw images $\mathbf{X}^t = \{\mathbf{x}_a^t, y_a^t\}_a^{N^t} \in \mathbb{R}^{p\times p\times B^t}$ from TD, data augmentation will be accomplished to obtain $\mathbf{X}_{\mathrm{I}}$ and $\mathbf{X}_{\mathrm{II}}$ through flipping and adding Gaussian noise, where $\mathbf{X}_{\mathrm{I}}$ and $\mathbf{X}_{\mathrm{II}}$ shared the same label y^t.

C. Metaknowledge Extraction Procedure

In the proposed DKD-FSL approach, the feature extractor initially embeds SD and TD data into a spatial–spectral embedding metric space, forming embedded features. Then, the categories of the query set are determined by measuring the distance between the support set and query set samples. The FSL for both SD and TD can be done in the same way. For instance, in the case of FSL for SD, $\mathbf{X}^s = \{\mathbf{X}_s^s, \mathbf{X}_q^s\}$ are initially fed into the DIR module; this process can be represented as

$$\mathbf{F}^s = \Phi\left(\mathbf{M}^{s/t}\left(\mathbf{X}^s\right)\right) \tag{1}$$

where $\Phi(\bullet)$ denotes the stands for DIR module.

Then, a multiscale feature extraction network $\psi(\bullet)$ is employed as the feature extractor to explore the metaknowledge within the data. This network primarily comprises two multiscale spectral–spatial feature extraction (MSFE) modules. An average pooling layer and three dilated convolution layers exist in an MSFE module, and the final multiscale spatial–spectral features are obtained by concatenating the outputs of the dilated convolutions. The output-embedded spatial–spectral features can be represented as

$$\mathbf{Z}^s = \psi\left(\mathbf{F}^s\right) \in \mathbb{R}^B \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

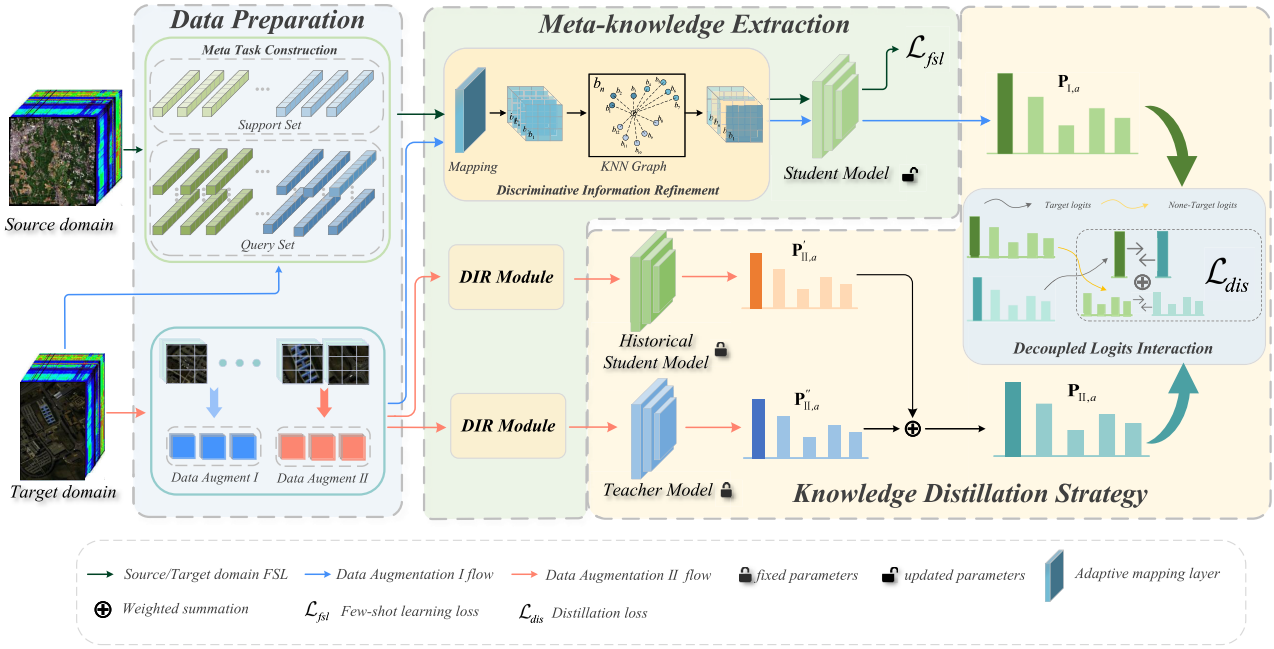


Fig. 1. Flowchart of the DKD-FSL method, mainly consists of three stages. In the data preparation stage, both the SD data and TD data will be structured into metatasks, and a part of the TD data will undergo data augmentation. During the metaknowledge extraction stage, the DIR module will preliminarily refine discriminative information within the data, while the student model will acquire metaknowledge from the data through the FSL strategy. In the KD phase, the teacher model will mimic the collaborative prediction of the teacher model and the historical student model. Meanwhile, the DLI module will decouple logit distillation into TCKD and NCKD to prevent the student model from simply imitating the probability distribution.

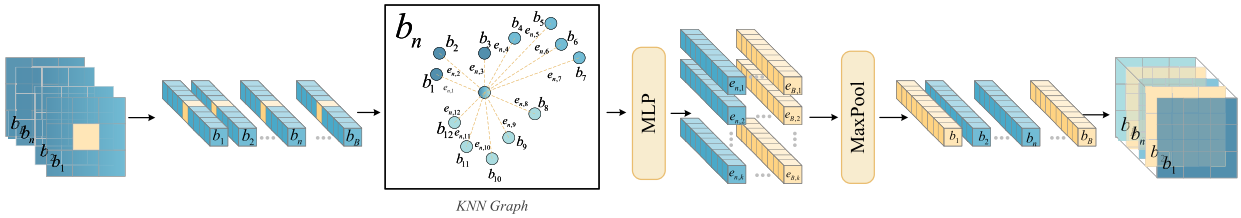


Fig. 2. Diagram of the DIR module.

where B represents the dimension after mapping by the DIR module.

For embedded features $\mathbf{Z}^s = \{\mathbf{Z}_s^s, \mathbf{Z}_q^s\}$, we treat the samples in the support set as prototypes for each class, calculate the distances between each query sample and the prototypes to determine classification results, and, finally, match them with the labels to obtain the FSL loss $\mathcal{L}_{\text{fsl}}^s$. In the proposed method, both Euclidean distance and class covariance measurements are utilized. Such a design provides a more comprehensive and flexible decision boundary compared to using only Euclidean distance metrics. Therefore, concerning the sample $\mathbf{z}_j^s$ in the query set, predicted outputs are provided in the form of probability based on two distinct distance

$$P_E\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right) = \text{Softmax}\left(-d_E\left(\mathbf{z}_j^s, \mu_j\right)\right) \tag{3}$$

$$P_M\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right) = \text{Softmax}\left(-d_M\left(\mathbf{z}_j^s, \mu_j\right)\right) \tag{4}$$

$$P\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right) = \left(P_E\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right) + P_M\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right)\right)/2 \tag{5}$$

where $d_E(\bullet)$ represents the Euclidean distance, $d_M(\bullet)$ represents the class covariance metric function, and μ_j denotes the prototype of category j, namely, the mean of all support samples. Based on $P(\hat{y}_j^s|\mathbf{z}_j^s)$, we can compute the FSL loss for a sample from a query set in SD as

$$\mathcal{L}_{\text{fsl}}^s\left(P\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right), y_j^s\right) = -y_j^s \log P\left(\hat{y}_j^s \middle| \mathbf{z}_j^s\right). \tag{6}$$

The FSL loss in TD can be calculated in a similar manner

$$\mathcal{L}_{\text{fsl}}^t\left(P\left(\hat{y}_j^t \middle| \mathbf{z}_j^t\right), y_j^t\right) = -y_j^t \log P\left(\hat{y}_j^t \middle| \mathbf{z}_j^t\right). \tag{7}$$

D. DIR Module

HSIs are known to contain numerous redundant bands, which means that the discriminative information is limited and varies in emphasis. If it is possible to enhance useful components and suppress irrelevant ones from redundant bands, it would not only facilitate more efficient acquisition of critical metaknowledge but also effectively reduce the negative impact of redundant information on metaknowledge transfer. Based on this idea, the DIR module was designed with the structure shown in Fig. 2.

First, in order to deal with the differences in spectral dimensions between $\mathcal{D}^s$ and $\mathcal{D}^t$ from different sensors, we employ an adaptive mapping layer to unify the data dimensions of $\mathbf{X}^s$ and $\mathbf{X}^t$. The adaptive mapping layer is composed of two 2-D convolutional layers with a total of B convolutional kernels of size $1 \times 1 \times c^{s/t}$, where $c^{s/t}$ denotes the dimensionality of the input data. The adaptive mapping layer can be denoted as $\mathbf{M}^{s/t}(\bullet)$. Then, a k-NN graph is used to explore relationships between spectral bands [53], [54].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

We first assume that the input is $\mathbf{x}_{\text{in}}^{s/t} = [\mathbf{b}_1^{s/t}, \mathbf{b}_2^{s/t}, \ldots, \mathbf{b}_{N_s}^{s/t}] \in \mathbb{R}^{B\times p^2}$, where $\mathbf{b}_n^{s/t}$ can be regarded as representation of single spectral band and $n = 1, 2, \ldots, N_s$. Subsequently, considering $\mathbf{b}_n^{s/t}$ as cluster prototypes, relationships between $\mathbf{b}_n^{s/t}$ and k most similar neighbors can be constructed using k-NN algorithm; these neighbors are $\mathbf{b}_{n,1}^{s/t}, \mathbf{b}_{n,2}^{s/t}, \ldots, \mathbf{b}_{n,k}^{s/t}$. Similar to this, the prototypes can be taken as different nodes of the graph, and edge features $\mathbf{e}_{(n,m)}^{s/t}$ between different prototypes are the edges in the graph. The edges can also be defined as

$$\mathbf{e}_{(n,m)}^{s/t} = f_\theta\left(\mathbf{b}_n^{s/t}, \mathbf{b}_{n,m}^{s/t} - \mathbf{b}_n^{s/t}\right) \tag{8}$$

where represents a nonlinear function, i.e., $f_\theta : \mathbb{R}^{p^2} \times \mathbb{R}^{p^2} \to \mathbb{R}^{p^2}$, while θ denotes a learnable parameter. After dynamic clustering, a complete graph is obtained, with its edges denoted as $\mathbf{e}_{(n,k)}^{s/t} \in \mathbb{R}^{k\times B\times p^2}$. Finally, symmetric aggregation of edges from each node is achieved by max pooling to obtain $\mathbf{x}_{\text{out}}^{s/t} = \text{MaxPool}(\mathbf{e}_{(n,k)}^{s/t}) \in \mathbb{R}^{B\times p^2}$.

E. KD Strategy

The core objective of CDFSL is to achieve extraction and debiasing for metaknowledge. To accomplish virtuous and sustainable knowledge interaction between SD and TD, the KD strategy is employed in the proposed DKD-FSL method. KD enables a student model to learn on its own while receiving guidance from the teacher for downstream tasks [55], [56]. With this characteristic, the student model can not only maintain efficient knowledge extraction in the proposed method but also learn the key knowledge of the TD and discard useless information according to the instruction of the teacher model.

Specifically, there are three models in the proposed framework: the student model $\mathcal{G}_S$, the teacher model $\mathcal{G}_T$, and the historical student model $\mathcal{G}_H$. Within this framework, the primary role of $\mathcal{G}_S$ is to leverage the FSL strategy to extract discriminative metaknowledge from abundant data. Simultaneously, $\mathcal{G}_T$ and $\mathcal{G}_S$ will make predictions for both weakly and strongly augmented image patches. The predictions of $\mathcal{G}_T$ serve as the optimization criterion for $\mathcal{G}_S$, aiming to guide the student model to focus on the TD data and comprehend its intrinsic characteristics. subsequently, $\mathcal{G}_H$ will collaborate with $\mathcal{G}_T$ to iteratively mitigate negative impact from bias of SD.

Specifically, the embedding features of $\mathbf{x}_{\text{I},a}$ and $\mathbf{x}_{\text{II},a}$ as weakly and strongly augmented versions of the same patch can be calculated using the following formula:

$$\mathbf{z}_{\text{I},a} = \mathcal{G}_T^o\left(\Phi\left(\mathbf{M}^t\left(\mathbf{x}_{\text{I},a}\right)\right)\right) \in \mathbb{R}^{1\times C} \tag{9}$$

$$\mathbf{z}'_{\text{II},a} = \mathcal{G}_S^o\left(\Phi\left(\mathbf{M}^t\left(\mathbf{x}_{\text{II},a}\right)\right)\right) \in \mathbb{R}^{1\times C} \tag{10}$$

where $\mathcal{G}_T^o$ and $\mathcal{G}_S^o$ share the same initial parameters trained on the SD data, and they are undergoing the oth training iteration. $\mathcal{G}_S^{o-1}$ will be saved as $\mathcal{G}_H^o$ and participate in training together with it. $\mathcal{G}_H^o$ will predict on the same input alongside $\mathcal{G}_T^o$

$$\mathbf{z}''_{\text{II},a} = \mathcal{G}_H^o\left(\Phi\left(\mathbf{M}^t\left(\mathbf{x}_{\text{II},a}\right)\right)\right) \in \mathbb{R}^{1\times C}. \tag{11}$$

The fusion of $\mathbf{z}'_{\text{II},a}$ and $\mathbf{z}''_{\text{II},a}$ can be formulated as

$$\mathbf{z}_{\text{II},a} = \alpha^o \mathbf{z}'_{\text{II},a} + \left(1 - \alpha^o\right)\mathbf{z}''_{\text{II},a} \tag{12}$$

where $\alpha = (o/T)$ represents an adaptive weight in the oth iteration, whose value is associated with the total number of iterations T. Finally, we can calculate the distillation loss for the oth iteration according to the formula of the Kullback–Leibler (KL) divergence

$$\mathcal{L}_{\text{dis}}^o = \begin{cases} \mathbb{KL}\left(\mathbf{P}_{\text{I},a} \| \mathbf{P}'_{\text{II},a}\right), & \text{if } o = 0 \\ \mathbb{KL}\left(\mathbf{P}_{\text{I},a} \| \mathbf{P}_{\text{II},a}\right), & \text{otherwise} \end{cases} \tag{13}$$

where $\mathbf{P}_{\text{I},a}$, $\mathbf{P}'_{\text{II},a}$, and $\mathbf{P}_{\text{II},a}$ are the corresponding outputs of the softmax function from $\mathbf{z}_{\text{I},a}$, $\mathbf{z}'_{\text{II},a}$, and $\mathbf{z}_{\text{II},a}$. As the iterations continue, the predictions of the historical student model for the weakly-augmented version $\mathbf{P}''_{\text{II},a}$ will dominate to correct for the bias in the student model's predictions $\mathbf{P}_{\text{I},a}$.

F. DLI Module

It is important to note that in the process of KD, student models will tend to simply mimic the probability distribution exhibited by the teacher model, thereby neglecting the exploration of deep semantic information. This is mainly due to the fact that in the conventional KD, the logit suffers from the coupling of task-related and data-intrinsic knowledge. To address this issue, we designed the DLI module in DKD-FSL to maintain a balance of multitype knowledge in the student model.

In detail, the prediction obtained by each model is a tensor consisting of C probability values, so $\mathbf{P}$ can be written as $\mathbf{P} = [p_1, p_2, \ldots, p_e, \ldots, p_C] \in \mathbb{R}^{1\times C}$. Here, p_e is the probability of eth class, which can be calculated by the softmax function

$$p_r = \frac{\exp(z_r)}{\sum_{d=1}^{C}\exp(z_d)} \tag{14}$$

where z_r is an element of $\mathbf{z} = [z_1, z_2, \ldots, z_e, \ldots, z_C]$. According to (14), we can also calculate the probability p_{ne} of all elements not belonging to the eth class, and p_{ne} can be denoted as

$$p_{ne} = 1 - p_r = \frac{\sum_{f=1, f\neq e}^{C}\exp\left(z_f\right)}{\sum_{d=1}^{C}\exp(z_d)}. \tag{15}$$

Meanwhile, we can divide $\mathbf{P}$ into two parts corresponding to whether it belongs to the eth class, namely, $\mathbf{P} = \mathbf{P}_e \cup \mathbf{P}_{ne}$, where $\mathbf{P}_{ne} = [\hat{p}_1, \hat{p}_2, \ldots, \hat{p}_{e-1}, \hat{p}_{e+1}, \ldots, \hat{p}_C]$. The element in $\mathbf{P}_{ne}$ can be calculated as (14)

$$\hat{p}_r = \frac{\exp(z_r)}{\sum_{f=1, f\neq e}^{C}\exp\left(z_f\right)}. \tag{16}$$

Using the same approach as (14) and (15), (13) can be rewritten as

$$\begin{aligned}
\mathcal{L}_{\text{dis}} &= \mathbb{KL}\left(\mathbf{P}^{\mathcal{T}} \| \mathbf{P}^{\mathcal{S}}\right) \\
&= \sum_{r=1}^{C} p_r^{\mathcal{T}} \log\frac{p_r^{\mathcal{T}}}{p_r^{\mathcal{S}}} \\
&= p_e^{\mathcal{T}} \log\frac{p_e^{\mathcal{T}}}{p_e^{\mathcal{S}}} + \sum_{r=1, r\neq e}^{C} p_r^{\mathcal{T}} \log\frac{p_r^{\mathcal{T}}}{p_r^{\mathcal{S}}} \\
&= p_e^{\mathcal{T}} \log\frac{p_e^{\mathcal{T}}}{p_e^{\mathcal{S}}} + p_{ne}^{\mathcal{T}} \sum_{r=1, r\neq e}^{C} \hat{p}_r^{\mathcal{T}} \left(\log\left(\frac{\hat{p}_r^{\mathcal{T}}}{\hat{p}_r^{\mathcal{S}}}\right) + \log\left(\frac{p_{ne}^{\mathcal{T}}}{p_{ne}^{\mathcal{S}}}\right)\right) \\
&= p_e^{\mathcal{T}} \log\frac{p_e^{\mathcal{T}}}{p_e^{\mathcal{S}}} + p_{ne}^{\mathcal{T}} \log\left(\frac{p_{ne}^{\mathcal{T}}}{p_{ne}^{\mathcal{S}}}\right) \\
&\quad + p_{ne}^{\mathcal{T}} \sum_{r=1, r\neq e}^{C} \hat{p}_r^{\mathcal{T}} \log\left(\frac{\hat{p}_r^{\mathcal{T}}}{\hat{p}_r^{\mathcal{S}}}\right).
\end{aligned} \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

In fact, (17) can be regarded as a combination of the expanded form of the KL divergence. Its concise form is

$$\mathcal{L}_{\text{dis}} = \mathbb{KL}\left(\mathbf{P}_d^{\mathcal{T}} \middle\| \mathbf{P}_d^{\mathcal{S}}\right) + \left(1 - p_e^{\mathcal{T}}\right)\mathbb{KL}\left(\mathbf{P}_{ne}^{\mathcal{T}} \middle\| \mathbf{P}_{ne}^{\mathcal{S}}\right). \tag{18}$$

Following the contents of (18), a reorganized distillation loss of conventional logit distillation is visually presented. These two components can be categorized as target class KD (TCKD) and nonclass KD (NCKD) based on whether they involve the probability of the target class [57].

Analyzing from the perspective of the CDFSL task, TCKD conveys task-related knowledge, which is strongly correlated with the confidence level of the target class and represents information related to the task decision. This does not imply that NCKD is ineffective, NCKD conveys data-intrinsic knowledge. It can be referred to as "dark knowledge," encompassing deep semantic representations of data attributes and distributions, which form the foundation of classification. TCKD and NCKD should ideally complement each other. However, NCKD is often suppressed by TCKD due to $1 - p_e^{\mathcal{T}}$ in traditional KD, leading to less effective "communication" between the student and teacher model.

In order to establish a balanced and comprehensive knowledge transfer paradigm toward CDFSL, hyperparameters α and β are used to weight the two components separately, and the distillation loss can be represented as

$$\begin{aligned}\mathcal{L}_{\text{dis}} &= \alpha\mathbb{KL}\left(\mathbf{P}_d^{\mathcal{T}} \middle\| \mathbf{P}_d^{\mathcal{S}}\right) + \beta\mathbb{KL}\left(\mathbf{P}_{ne}^{\mathcal{T}} \middle\| \mathbf{P}_{ne}^{\mathcal{S}}\right) \\ &= \alpha\text{TCKD} + \beta\text{NCKD}.\end{aligned} \tag{19}$$

Finally, the overall loss of DKD-FSL is expressed as the sum of the two parts

$$\mathcal{L} = \mathcal{L}_{\text{fsl}}^{s/t} + \lambda\mathcal{L}_{\text{dis}} \tag{20}$$

where λ is the regularization parameter adjusting the balance between FSL learning and KD.

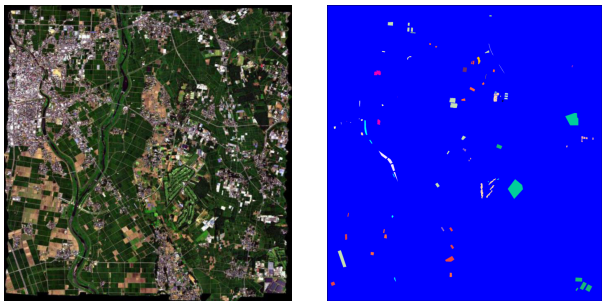
In the metatesting stage, the student model $\mathcal{G}_S$ optimized through the KD strategy will be employed to extract discriminative features of the unlabeled samples, which can then be classified by the k-NN classifier to obtain predictions.

III. Experimental Results and Analysis

In this section, the experimental datasets and settings will be briefly introduced first. Then, the experimental results of comparison with mainstream methods will be presented and analyzed. Subsequently, the results of ablation experiments and feature visualization will be provided. The final part will discuss the parameter sensitivity related to the DKD-FSL method.

A. Experimental Datasets

The experimental datasets utilized in this article can be classified into SD datasets and TD datasets, including Chikusei [58], Pavia University (PU), DongTing Lake (DTL) [59], and Yancheng (YC) Wetlands [60]. Chikusei was selected as the SD dataset due to the sufficient annotated samples; the other datasets were employed as the TD dataset.



(a) (b)

Fig. 3. (a) Pseudocolor image and (b) ground-truth map of Chikusei dataset.

TABLE I
Land Cover Classes and the Number of Samples for the Chikusei Dataset

No.	Color	Land cover type	Samples
1		Water	2345
2		Bare soil (school)	2859
3		Bare soil (park)	236
4		Bare soil (farmland)	48525
5		Natural plants	4297
6		Weeds in farmland	1108
7		Forest	20516
8		Grass	6515
9		Rice filed (grown)	13369
10		Rice filed (first stage)	1268
11		Row crops	5961
12		Plastic house	2193
13		Manmade (non-dark)	1220
14		Manmade (dark)	7664
15		Manmade (blue)	431
16		Manmade (red)	222
17		Manmade grass	1040
18		Asphalt	801
19		Paved ground	145
		Total	77592

1) SD Dataset: The Chikusei dataset was captured by the airborne imaging device hyperspectral visible/near-infrared C-series (Hyperspec-VNIR-C) sensor on July 29, 2014, under favorable weather conditions. The dataset consists of images measuring 2517 × 2335 pixels with a spatial resolution of 2.5 m, captured in Chikusei, Japan. It encompasses a spectral range from 363 to 1018 nm and is divided into 128 bands. The dataset contains 19 main land-cover types and 77 592 labeled samples. The specific numbers for each category are provided in Table I. In addition, the pseudocolor image and ground-truth map are displayed in Fig. 3.

2) TD Dataset: The PU, DTL, and YC datasets were selected as the TD datasets. Specific information regarding the land cover types in the TD datasets is provided in Tables II–IV.

The PU dataset was captured in 2003 with a reflective optics spectrographic imaging system (ROSIS-03) sensor. This dataset, with a spatial resolution of 1.3 m, covered 610 × 340 pixels over the area of PU in Italy. The dataset has a total

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

TABLE II
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE PU DATASET

No.	Color	Land cover type	Samples
1		Asphalt	6631
2		Meadow	18549
3		Gravel	2099
4		Trees	3064
5		Sheets	1345
6		Bare soil	5029
7		Bitumen	1330
8		Bricks	3682
9		Shadow	947
		Total	42776

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE DTL DATASET

No.	Color	Land cover type	Samples
1		Lake water	1120
2		Tidal flat	736
3		Meadow	877
4		Reed	1006
5		Willow	644
6		Carex	433
		Total	4816

TABLE IV
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE YC WETLANDS DATASET

No.	Color	Land cover type	Samples
1		Building	451
2		Suaeda	269
3		River	214
4		Beach	331
5		Bare land	235
6		Paddy field	3026
7		Idle land	650
8		Fish pond	920
9		Sea	2076
10		Dry land	145
11		Salt pan	901
12		Reed	244
13		Shallow sea	1558
14		Pond	68
15		Spartina alterniflora	306
16		Ditch	328
		Total	11722

of 103 available spectral bands. The pseudocolor images of the PU dataset are presented in Fig. 4(a).

The DTL dataset was photographed in the DTL basin in Hunan Province, China, via the GF-5 satellite in 2019. The dataset has a spatial resolution of 30 m and consists of 456 × 352 pixels. It has 310 available bands after removing 20 broken bands. It includes a total of 4816 annotated samples and encompasses six categories. The pseudocolor image of the DTL dataset is depicted in Fig. 5(a).

The YC dataset was captured in 2020 by the ZY1-02D satellite in Jiangsu Province, China, using a visible and near-infrared camera and an advanced hyperspectral imager (AHSI) sensor. With a spatial resolution of 30 m, the YC dataset encompasses a range of 1398 × 942 pixels and 147 bands. This dataset comprises a total of 11 722 annotated samples across 16 categories. The pseudocolor image and the ground-truth map of the YC dataset are illustrated in Fig. 6(a) and (b).

B. Experimental Settings

The DKD-FSL method is implemented by using the Python and PyTorch platforms on Ubuntu 20.04. All relevant experiments in this section were conducted on the same hardware platform, which includes NVIDIA Geforce RTX-3090, Intel Silver 4210R CPU, and 56-GB memory. In the metatraining, only classes with annotated samples greater than 200 are retained in the Chikusei dataset, and 200 samples per class are randomly selected to participate in the FSL for SD. In the TD dataset, five labeled samples from each class are selected to participate in the FSL for TD. The construction of the metatask follows the requirements of the C-way K-shot tasks, with specific values depending on the TD. The numbers of the support set and the query set are 1 and 19, respectively.

All samples will be involved in training as hyperspectral patches, the patch size is 9, and the number of bands will be embedded to $B = 100$. The adaptive mapping module and feature extractor adopt the same settings, as described in [43], and the Adam optimizer with a learning rate of 0.01 is also used for training. The training number of episodes is set to 5000 to fully mine the metaknowledge within the data. In the DKD-FSL, the student model and the teacher model with the same structure require a simple pretraining to obtain the initial parameters. During the pretraining stage, the model employs an episodic learning strategy with the help of metatasks for short-term training. Its specific setup is the same as the formal training. The number of pretraining episodes is set to 2000, and no additional training samples are introduced. After pretraining, the parameters of the model are retained and used for formal training. To provide intuitive measurement criteria, all experiments involved adopting overall accuracy (OA), average accuracy (AA), and Kappa coefficient as unified evaluation metrics.

C. Classification Results of Different Methods

To assess the effectiveness of the proposed method, several mainstream methods are applied as comparison benchmarks, including SVM, SSRN [19], SSFTT [22], CDFSL [40], CMFSL [46], Gia-FSL [41], and DMCM [43]. Among them, SVM is a traditional ML-based method, SSRN and SSFTT represent the DL-based spectral–spatial methods, and the rest of them are the latest methods based on the FSL strategy. The SDs for CDFSL, CMFSL, Gia-FSL, and DMCM are all in the Chikusei dataset. The CMFSL method employs optimal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

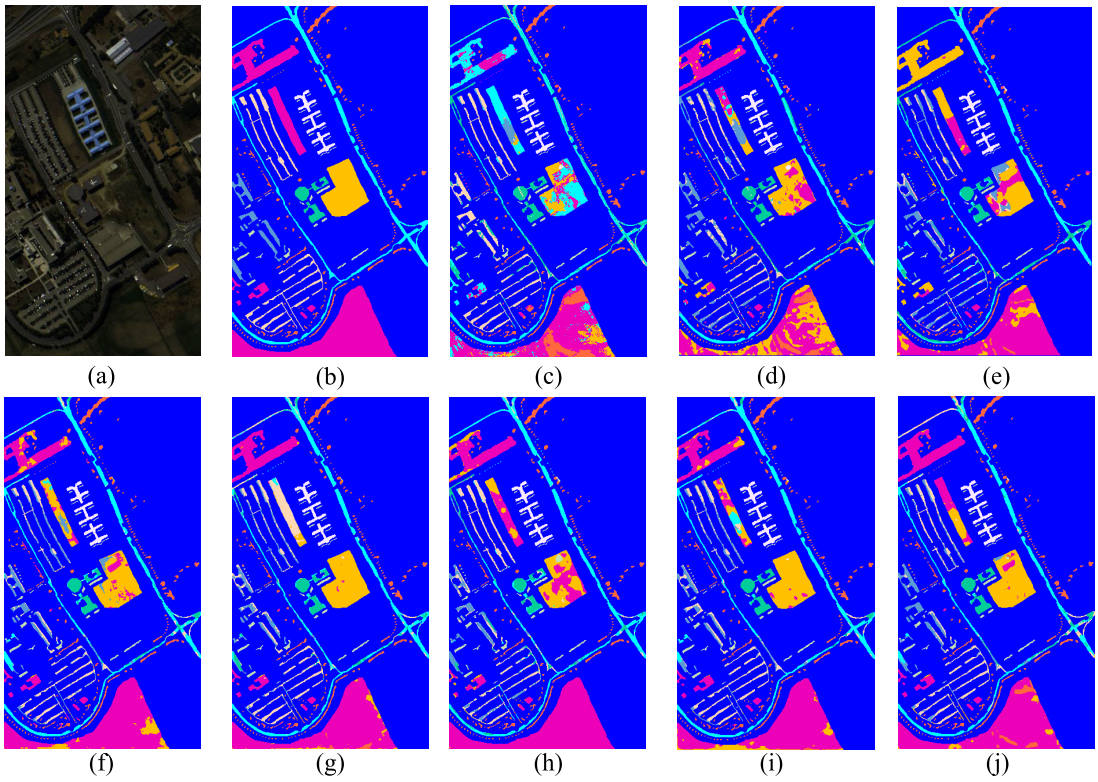


Fig. 4. Classification maps of different methods on the PU dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) SVM (56.35%). (d) SSRN (69.05%). (e) SSFTT (%). (f) CDFSL (81.42%). (g) CMFSL (80.09%). (h) Gia-FSL (81.69%). (i) DMCM (84.41%). (j) DKD-FSL (87.66%).

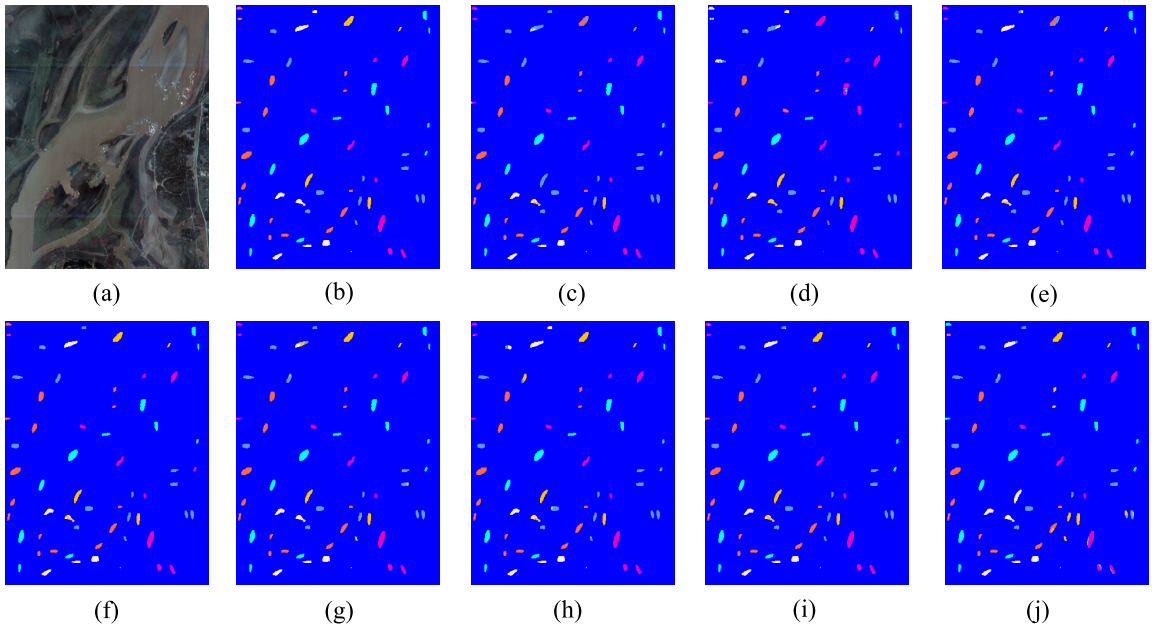


Fig. 5. Classification maps of different methods on the DTL dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) SVM (86.53%). (d) SSRN (88.03%). (e) SSFTT (92.01%). (f) CDFSL (94.87%). (g) CMFSL (94.65%). (h) Gia-FSL (94.90%). (i) DMCM (94.75%). (j) DKD-FSL (95.26%).

clustering (OCBS) to process the raw SD data and the TD data to align the number of bands. Each of the three TD datasets had five labeled samples per class, which were used in the training process. All other specific hyperparameters involved are set according to the default settings introduced in the corresponding articles. In addition, to mitigate the impact of randomness on the results, all methods are subjected to ten independent repeated experiments with the same set of random seeds. All experimental results are presented in Tables V–VII with mean accuracy and standard deviation. The highest accuracy for each evaluation metric is highlighted in bold.

1) Classification Results on the PU Dataset: Table V provides the classification results of various methods on the PU dataset, and the classification maps are also depicted in Fig. 4. It is evident that the traditional HSIC methods undergo

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

TABLE V
CLASSIFICATION ACCURACIES OF METHODS ON THE PU DATASET

Class	SVM	SSRN	SSFTT	DCFSL	CMFSL	Gia-FSL	DMCM	DKD-FSL
1	49.96±8.42	83.33±10.07	63.88±8.82	75.63±13.42	75.89±11.49	77.99±9.97	84.15±5.02	**85.14±4.51**
2	49.96±21.57	55.14±17.36	59.80±18.92	86.12±6.41	80.34±11.20	86.96±6.56	83.69±13.50	**90.45±4.83**
3	25.13±19.09	69.81±9.03	63.97±16.89	65.83±9.82	61.15±17.41	58.37±10.88	**71.67±11.08**	69.69±12.99
4	85.78±10.58	90.76±6.56	89.97±10.19	92.79±3.91	92.01±4.15	93.13±3.25	93.17±4.32	**94.21±2.40**
5	95.05±8.61	98.71±1.47	**99.70±0.40**	99.31±0.63	99.44±0.62	98.75±2.03	98.68±2.69	99.15±0.96
6	42.50±22.07	68.57±17.85	74.41±11.77	76.76±8.24	81.18±9.67	71.99±10.75	78.05±12.86	**81.20±11.46**
7	93.94±2.85	87.40±14.31	85.15±12.63	80.54±8.69	83.25±4.79	76.87±8.30	**90.69±5.43**	86.50±8.21
8	73.69±12.94	70.82±10.02	70.09±13.93	63.12±13.20	72.99±12.24	69.97±12.82	**86.02±9.80**	85.18±8.34
9	**99.83±0.10**	99.60±0.79	96.25±3.96	99.04±0.96	98.31±2.15	98.09±3.41	99.10±1.47	98.69±1.49
OA(%)	56.35±7.56	69.05±5.38	68.24±7.71	81.42±2.21	80.09±4.28	81.69±3.18	84.41±5.24	**87.66±2.41**
AA(%)	68.43±2.39	80.46±1.82	78.14±3.08	82.12±1.14	82.72±1.85	81.34±1.83	87.24±1.60	**87.80±2.25**
Kappa	47.11±6.63	62.09±5.34	61.15±5.34	75.95±2.53	74.60±4.89	76.21±3.77	79.98±5.91	**83.88±3.09**

TABLE VI
CLASSIFICATION ACCURACIES OF METHODS ON THE DTL DATASET

Class	SVM	SSRN	SSFTT	DCFSL	CMFSL	Gia-FSL	DMCM	DKD-FSL
1	**99.82±0.00**	97.64±5.88	99.72±0.34	99.13±1.00	99.63±0.56	98.87±1.53	99.74±0.47	99.04±1.30
2	**99.99±0.04**	99.82±0.33	59.80±18.92	99.95±0.13	98.06±5.83	99.26±1.00	**99.99±0.04**	99.85±0.26
3	82.22±16.36	83.19±12.13	87.16±14.93	**92.34±4.93**	91.74±8.12	91.80±5.30	91.87±7.13	91.41±6.10
4	91.49±2.76	83.45±10.46	89.58±4.71	91.84±3.40	90.63±4.56	92.73±3.67	91.40±3.05	**92.30±3.48**
5	77.97±9.06	87.32±11.18	90.36±6.61	**92.47±6.06**	92.24±5.91	91.13±5.03	91.75±7.55	92.14±6.29
6	38.86±16.37	80.77±15.37	76.64±13.04	91.00±12.44	94.84±9.66	94.18±9.76	91.05±12.27	**97.03±7.92**
OA(%)	86.53±1.80	88.03±3.08	92.01±2.19	94.87±1.82	94.65±1.48	94.90±1.60	94.75±1.91	**95.26±1.55**
AA(%)	81.72±1.79	87.11±3.40	90.55±2.13	94.45±2.55	94.52±1.78	94.66±2.18	94.29±2.65	**95.30±2.01**
Kappa	83.53±2.14	85.46±3.72	90.27±2.63	93.76±2.21	93.49±1.79	93.79±1.95	93.61±2.31	**94.23±1.90**

TABLE VII
CLASSIFICATION ACCURACIES OF METHODS ON THE YC WETLANDS DATASET

Class	SVM	SSRN	SSFTT	DCFSL	CMFSL	Gia-FSL	DMCM	DKD-FSL
1	49.17±12.17	71.43±17.46	75.81±13.29	81.77±10.65	**92.60±6.91**	82.89±11.55	79.35±14.06	85.58±8.83
2	87.39±1.94	87.42±8.16	90.42±6.33	**94.77±4.24**	90.45±9.89	88.86±6.92	90.80±7.34	92.35±5.93
3	61.82±16.00	91.29±7.77	83.54±12.95	89.57±9.80	90.72±11.11	82.54±20.37	91.77±7.17	**91.77±7.02**
4	49.97±20.84	85.98±10.95	33.25±27.15	**95.74±2.54**	95.34±6.62	94.51±5.00	94.88±7.10	91.75±10.03
5	**100.00±0.00**	98.57±0.39	98.57±0.28	99.65±0.17	99.43±0.39	99.61±0.36	99.22±0.58	99.43±0.39
6	94.84±3.38	97.68±4.04	**99.76±0.45**	95.12±6.30	93.53±8.26	93.94±7.43	95.25±5.74	98.10±3.46
7	93.10±0.80	95.91±2.67	94.54±5.44	96.37±3.03	95.81±2.39	93.55±3.93	94.67±3.33	**96.43±3.08**
8	19.19±19.49	**77.67±5.35**	44.30±18.96	71.77±8.51	69.89±14.52	72.67±10.39	76.60±4.48	75.54±8.01
9	86.71±4.85	86.41±20.42	88.87±14.39	97.63±5.51	95.13±9.19	97.24±6.51	95.91±6.48	**97.99±5.85**
10	61.36±21.55	90.71±7.93	85.36±6.96	95.36±5.32	**96.21±5.30**	95.93±6.23	94.93±6.96	93.43±7.33
11	45.39±27.32	90.10±6.91	78.49±21.12	92.62±5.28	88.21±6.37	**92.75±4.18**	92.14±5.89	90.03±9.67
12	**99.33±0.98**	96.78±4.69	92.76±5.12	98.28±2.20	99.33±0.82	98.58±1.57	99.08±0.99	98.70±1.90
13	83.68±16.08	78.11±27.76	87.68±18.95	87.35±6.47	85.64±8.19	**89.31±6.47**	85.58±5.96	87.80±7.02
14	93.02±10.53	96.19±4.88	74.76±29.99	95.08±5.61	98.10±2.33	94.13±4.87	**98.25±2.60**	94.60±11.07
15	67.54±9.73	85.91±6.35	90.56±10.05	**90.80±10.39**	89.17±10.70	88.70±10.95	90.70±10.21	86.15±12.52
16	18.70±10.99	65.45±9.45	68.27±11.96	76.72±10.56	**85.02±9.70**	72.35±12.83	78.36±12.42	72.60±11.45
OA(%)	75.25±4.54	87.84±6.49	85.06±3.40	91.51±2.12	90.46±3.08	90.89±2.71	91.13±2.55	**92.27±2.24**
AA(%)	69.45±3.40	87.23±4.07	80.43±3.76	91.16±1.32	**91.53±1.28**	89.84±2.55	91.09±2.13	90.77±2.14
Kappa	71.63±5.02	86.04±7.33	82.72±3.87	90.21±2.42	89.03±3.50	89.51±3.09	89.79±2.91	**91.06±2.58**

significant performance degradation with insufficient number of samples, while the methods based on FSL, especially the DKD-FSL method, exhibit very competitive classification performance.

Evaluated according to the OA, AA, and Kappa coefficient metrics, DKD-FSL outperforms all comparisons, with a lead of 3.25%, 0.56%, and 3.9% over the second-best method, respectively. For the class 2 "Meadow," which has the largest number of samples, DKD-FSL achieved a considerable increase in accuracy and is the only method that correctly classifies more than 90% of the samples. In addition, for class 1 "Asphalt," due to its widely distributed samples and greater spectral diversity, most methods suffered significant misclassifications. However, the DKD-FSL method still maintains the highest accuracy and good stability in this class. Such an excellent and stable performance may stem from the sensitivity of the teacher model to key features in KD, which also implies that the dynamic debiasing process designed in the DKD-FSL method is effective.

2) Classification Results on the DTL Dataset: The classification maps on the DTL dataset of each method are illustrated in Fig. 5, while the specific classification results are provided in Table VI. The DKD-FSL method still shows the best and most stable classification performance on the DTL dataset,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

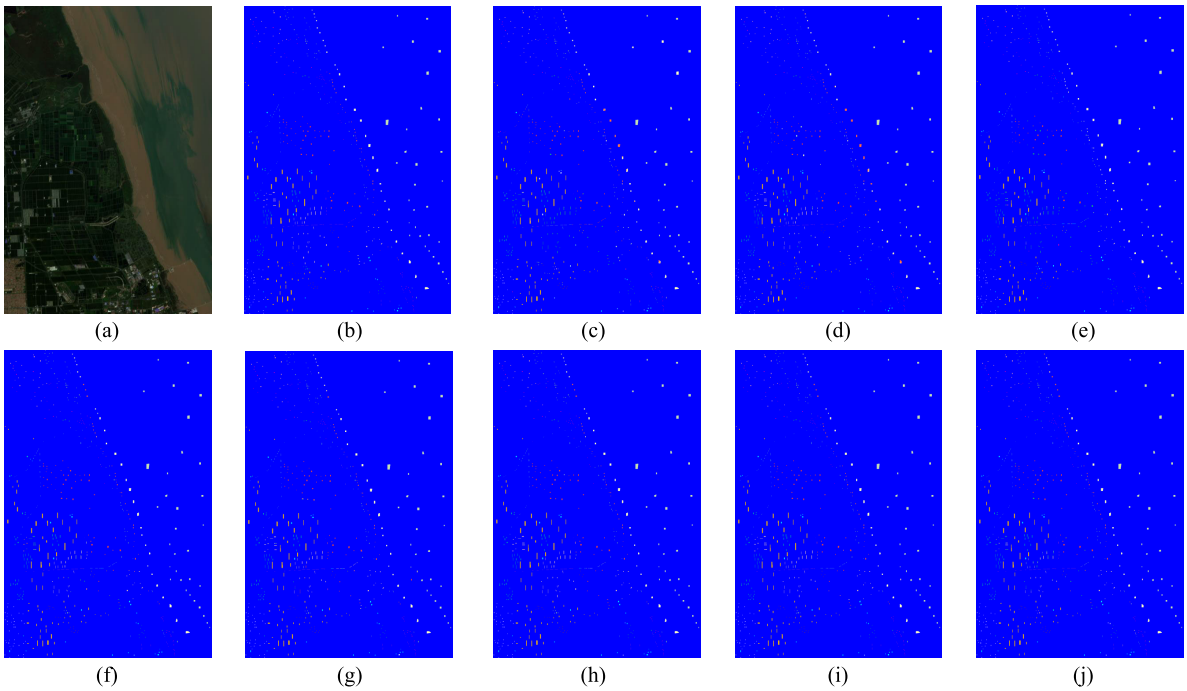


Fig. 6. Classification maps of different methods on the YC Wetlands dataset. (a) Pseudocolor image. (b) Ground-truth map. (c) SVM (75.25%). (d) SSRN (87.84%). (e) SSFTT (85.06%). (f) CDFSL (91.51%). (g) CMFSL (90.46%). (h) Gia-FSL (90.89%). (i) DMCM (91.13%). (j) DKD-FSL (92.27%).

achieving OA, AA, and Kappa coefficients of 95.26%, 95.30%, and 94.23% respectively.

Due to the concentrated distribution of samples within classes in the DTL dataset and the obvious gap between classes, all methods maintain a decent level of classification accuracy even with insufficient samples, and the OA for all methods remains above 85%. Especially for class 1 "Lake water" and class 2 "Tidal flat," all methods achieve near-perfect classification performance. Although DKD-FSL did not achieve a significant improvement in OA, the proposed method indicates a more balanced and competitive performance when considering individual classes. In the class 6 "Carex," where other methods perform poorly, DKD-FSL outperforms the second-best method by 2.19%. Due to the predominance of wetland land cover categories in the DTL dataset, which significantly differs from the SD dataset, the DKD-FSL method still efficiently classifies each category. This phenomenon proves that the KD-based metaknowledge translation is simple and effective in retaining discriminative information.

3) Classification Results on the YC Dataset: Fig. 6 and Table VII show the classification maps and detailed classification results of the proposed DKD-FSL method on the YC dataset, respectively. In comparison with OA and Kappa, the DKD-FSL method maintains the optimal overall classification performance with 92.27% and 91.06%.

It should be noted that although the proposed method does not attain the ideal classification in the comparison of AA, it achieves optimal or suboptimal results for class 6 "Paddy field," class 9 "Sea," and class 13 "Shallow sea," which have the largest number of samples in the YC dataset. This phenomenon indicates that the proposed method is capable of accurately modeling more diverse classes even with limited prior knowledge. From another perspective, the balanced classification results across various classes show the comprehensive mastery of key information by the DKD-FSL method, which may stem from the method relying on domain-general discriminative information rather than prior knowledge of a specific class.

4) Classification Results With Different Numbers of Labeled Samples: In order to investigate the effect of the prior information provided by the annotated samples on the classification performance, the experimental results with different numbers of annotated samples L are depicted in Fig. 7, where $L = \{1, 2, 3, 4, 5\}$. As expected, a decrease in L leads to varying degrees of performance degradation across all methods, indicating that the supervised information is still one of the main factors affecting the accuracy. It is noteworthy that the DKD-FSL method consistently maintains optimal classification accuracy and higher stability across the three datasets in all cases.

D. Ablation Study and Visualization

In order to intuitively and detailedly exhibit the performance enhancement brought by DKD-FSL, an ablation study on different modules was conducted in this section. In addition, a 2-D visualization comparison between the original data and the embedded features is provided.

1) Sequential Ablation Study of DKD-FSL Method: To further validate the effectiveness of the proposed DKD-FSL method and explore the role of the core modules, sequential ablation experiments are employed in this study. With the baseline model, three modules including the DIR module, the KD strategy, and the DLI module are added sequentially to complete the classification under the few-sample condition. The specific experimental results of different variants with five labeled samples are presented in Table VIII.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

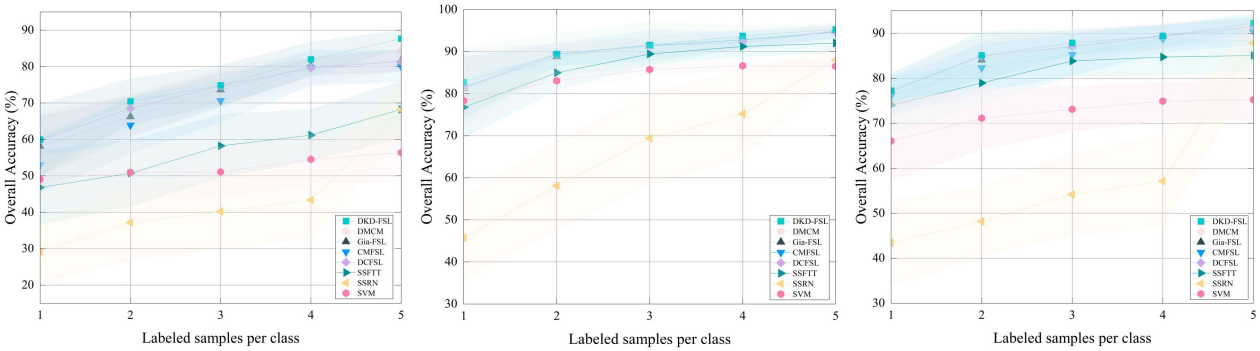


Fig. 7. Classification accuracy (OA) of various methods with different numbers of labeled samples on three datasets. (a) PU dataset. (b) DTL dataset. (c) YC dataset.

TABLE VIII
SEQUENTIAL ABLATION COMPARISON ON THREE DATASETS

Dataset	DIR	KD	DLI	OA(%)	AA(%)	Kappa
Pavia University	×	×	×	84.30±4.65	86.64±2.68	79.71±5.36
	✓	×	×	84.97±4.47	86.98±2.17	80.62±5.21
	✓	✓	×	86.66±4.06	87.20±1.65	82.70±4.84
	✓	✓	✓	**87.66±2.41**	**87.80±2.25**	**83.88±3.09**
Dongting Lake	×	×	×	93.89±1.87	93.28±2.67	92.56±2.28
	✓	×	×	94.51±2.16	94.02±2.89	93.32±2.63
	✓	✓	×	94.78±2.16	94.49±2.83	93.65±2.63
	✓	✓	✓	**95.26±1.55**	**95.30±2.01**	**94.23±1.90**
Yancheng Wetlands	×	×	×	90.70±2.51	89.93±2.71	89.28±2.85
	✓	×	×	91.23±2.17	90.38±1.87	89.88±2.47
	✓	✓	×	91.81±1.91	90.61±1.78	90.53±2.18
	✓	✓	✓	**92.27±2.24**	**90.77±2.14**	**91.06±2.58**

With the addition of the DIR module, the model achieves performance gains of 0.67%, 0.62%, and 0.53% compared to the benchmark on the three datasets, respectively. Similar improvements are also observed in terms of AA and Kappa. It is evident that the DIR module facilitates the model in summarizing discriminative knowledge.

Subsequently, the FSL strategy was embedded into a conventional logit distillation framework, which also incorporates a historical student to achieve iterative debiasing. This framework resulted in improvements of 1.69%, 0.27%, and 0.58% in terms of OA across the three datasets. This phenomenon suggests that the knowledge interaction between the teacher model and the student model serves as a beneficial mechanism for metaknowledge transfer between the SD and TD. More importantly, this global framework facilitates the acquiring unbiased metaknowledge from the SD and its coherent application for the classification task, without the need for additional DA modules.

Finally, the introduction of the DLI module has led to an OA of 87.66%, 95.26%, and 92.27% across the three datasets. This module clearly mitigates the negative impact traditionally associated with vague knowledge by balancing the decoupled components. Overall, the three modules proposed in the article are coordinated and mutually reinforcing with the FSL strategy and effectively enhancing the efficiency and performance of FSL learning with limited annotated samples.

2) Feature Visualization of DKD-FSL Method: In order to visually analyze the classification performance of DKD-FSL, we employ the t-SNE algorithm to reduce the dimensionality of the original data from various datasets, as well as the high-dimensional features extracted by DKD-FSL, enabling visualization in a 2-D space. The original distributions of the PU, DTL, and YC datasets are depicted in Fig. 9(a)–(c), while the distributions of features extracted by DKD-FSL are illustrated in Fig. 9(e)–(g).

In the PU dataset, the original distribution appears relatively scattered, with intermingled data from different categories, making it difficult to distinguish. However, DKD-FSL conspicuously embeds data of the same class in more concentrated positions, thereby rendering the classification boundaries easily discernible. Particularly notable are class 3 "Gravel," class 7 "Bitumen," and class 8 "Bricks" within the rectangular box in Fig. 9(d). The proposed method clearly separates them into three distinct clusters.

In the DTL dataset, the original data distribution is centered around several clusters, yet the distances between these centers of the same category are considerable. However, features extracted by DKD-FSL noticeably reduce the intracluster distances and effectively minimize the distances between centers. Notably, the original data of class 5 "Willow" exhibit two widely separated clusters. However, as shown in Fig. 9(e) within the rectangular box, the data of this class now possess a unified clustering center.

In the original data of the YC dataset, the features are distributed sparsely, resulting in small interclass distances among data clusters and low discriminability between multiple categories, such as class 3 "River" and class 16 "Ditch." However, in the feature distribution depicted in Fig. 9(g), these two categories exhibit clear boundaries. Furthermore,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

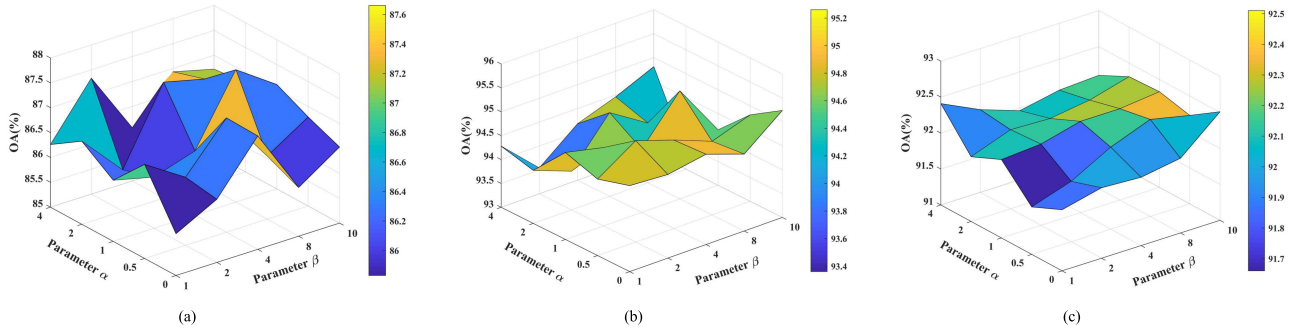


Fig. 8. Parameter tuning of α and β for the proposed DKD-FSL on three datasets. (a) PU. (b) DTL. (c) YC.

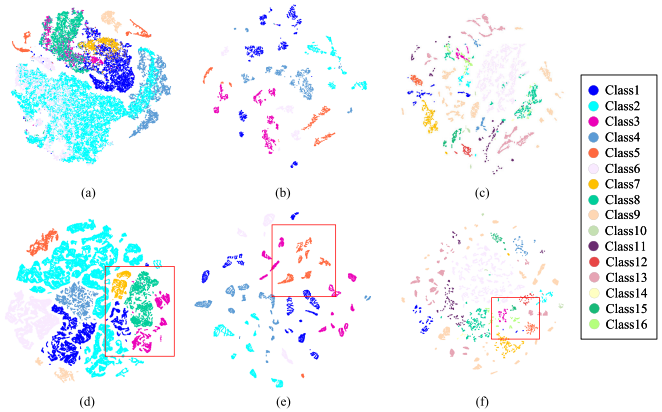


Fig. 9. t-SNE visualization of the original data and embedding features by DKD-FSL method on three datasets. (a) Original data of PU. (b) Original data of DTL. (c) Original data of YC. (d) Features of PU. (e) Features of DTL. (f) Features of YC.

it is easy to observe that the remaining categories tend to become more separated and independent clusters with greater distinctiveness.

E. Parameter Analysis

This section focuses on analyzing the two tunable parameters α and β in the DLI module. These parameters are primarily responsible for maintaining balanced knowledge transfer during distillation training in the DKD-FSL method. As independent weights affecting the two decoupled components, they effectively prevent the suppression of any component in an unknown situation, thereby avoiding performance degradation. To further evaluate the sensitivity of the proposed method to the two parameters, a hyperparameter sensitivity analysis was conducted on three datasets. The value ranges α and β were set to $\{0, 0.5, 1, 2, 4\}$ and $\{1, 2, 4, 8, 10\}$, respectively.

Fig. 8 illustrates the accuracy trends of the DKD-FSL method across three datasets with different combinations of α and β. When the values of α and β taken uneven values, such as $(\alpha = 4, \beta = 1)$ or $(\alpha = 0, \beta = 10)$, there is a noticeable decline in OA. Moreover, the closer these values are to such extreme settings, the greater the instability in classification performance. This may be attributed to a specific type of knowledge dominating the distillation training in such circumstances, thus overlooking other metaknowledge. In contrast, the trend of accuracy variation is slower around more balanced values.

IV. Conclusion

In this article, a CDFSL method based on decoupled KD (DKD-FSL) was proposed for HSIC. The proposed method aims to achieve unbiased metaknowledge extraction in a more coordinated and efficient way through a KD strategy. By facilitating knowledge interaction between multiple models, the approach guides the student model to acquire more discriminative metaknowledge and iteratively eliminates bias from SD. Meanwhile, to avoid incomplete knowledge transfer caused by coupled knowledge in logit distillation, the process is decoupled into two independent processes by the DLI module. These two components separately control the interaction of task-related knowledge and data-intrinsic knowledge between models, implementing metaknowledge communication in a more balanced and transparent manner. In addition, a DRI module is introduced into this framework, which enhances the distinctiveness of the discriminative information embedded in redundant bands of HSI. Comprehensive and extensive experiments and comparisons were conducted on three datasets, and the experimental results indicated the superior classification performance of the DKD-FSL method under conditions of limited samples.

As far as we know, although our proposed method is the first try to introduce a KD strategy in CDFSL, it is important to acknowledge the limitations of our work. First, the use of multiple models for training and inference in this framework results in a substantial increase in computational time. In addition, due to constraints related to computational costs, we have not been able to explore various scenarios where there are architectural differences between the teacher and student models. Our future work will focus on addressing these issues.

References

[1] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[2] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[3] R. Thoreau, V. Achard, L. Risser, B. Berthelot, and X. Briottet, "Active learning for hyperspectral image classification: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 256–278, Sep. 2022.

[4] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

[5] M. Imani and H. Ghassemian, "An overview on spectral and spatial information fusion for hyperspectral image classification: Current trends and challenges," *Inf. Fusion*, vol. 59, pp. 59–83, Jul. 2020.

[6] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[7] G. Taskin, H. Kaya, and L. Bruzzone, "Feature selection based on high dimensional model representation for hyperspectral images," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2918–2928, Jun. 2017.

[8] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[9] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[10] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[11] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified k-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jul. 2008.

[12] Y. Ren, Y. Zhang, W. Wei, and L. Li, "A spectral–spatial hyperspectral data classification approach using random forest with label constraints," in *Proc. IEEE Workshop Electron., Comput. Appl.*, May 2014, pp. 344–347.

[13] Q. Du and H. Ren, "Real-time constrained linear discriminant analysis to target detection and classification in hyperspectral imagery," *Pattern Recognit.*, vol. 36, no. 1, pp. 1–12, Jan. 2003.

[14] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[15] P. Wang, H. Yao, C. Li, G. Zhang, and H. Leung, "Multiresolution analysis based on dual-scale regression for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2021.

[16] P. Wang, L. Wang, H. Leung, and G. Zhang, "Super-resolution mapping based on spatial–spectral correlation for spectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2256–2268, Mar. 2021.

[17] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[18] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[19] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2017.

[20] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, Nov. 2023.

[21] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2020.

[22] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.

[23] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[24] W. Li et al., "Data augmentation for hyperspectral image classification with deep CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 593–597, Apr. 2018.

[25] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Hyperspectral image classification using random occlusion data augmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 11, pp. 1751–1755, Nov. 2019.

[26] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[27] B. Cui, X.-M. Dong, Q. Zhan, J. Peng, and W. Sun, "LiteDepthwiseNet: A lightweight network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502915.

[28] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[29] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.

[30] L. Zhao, W. Luo, Q. Liao, S. Chen, and J. Wu, "Hyperspectral image classification with contrastive self-supervised learning under limited labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[32] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.

[33] M. Cai et al., "Mind the gap: Multilevel unsupervised domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5520014.

[34] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.

[35] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.

[36] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.

[37] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.

[38] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2018.

[39] B. Guo, T. Liu, X. Zhang, and Y. Gu, "Few-shot multispectral-hyperspectral image collaborative classification with feature distribution enhancement and subdomain alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505117.

[40] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[41] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1912–1925, Feb. 2024.

[42] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.

[43] L. Hu, W. He, L. Zhang, and H. Zhang, "Cross-domain meta-learning under dual-adjustment mode for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526416.

[44] F. Liu, W. Gao, J. Liu, X. Tang, and L. Xiao, "Adversarial domain alignment with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525720.

[45] Q. Liu, J. Peng, N. Chen, W. Sun, Y. Ning, and Q. Du, "Category-specific prototype self-refinement contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5524416.

[46] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[47] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[48] Y. Song, T. Wang, P. Cai, S. K. Mondal, and J. P. Sahoo, "A comprehensive survey of few-shot learning: Evolution, applications, challenges, and opportunities," *ACM Comput. Surv.*, vol. 55, no. 13, pp. 1–40, Dec. 2023.

[49] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[50] S. Sun, W. Ren, J. Li, R. Wang, and X. Cao, "Logit standardization in knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 15731–15740.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.

[51] G. Xu, Z. Liu, X. Li, and C. C. Loy, "Knowledge distillation meets self-supervision," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Springer, 2020, pp. 588–604, doi: 10.1007/978-3-030-58545-7_34.

[52] K. Xu, L. Rui, Y. Li, and L. Gu, "Feature normalized knowledge distillation for image classification," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2020, pp. 664–680, doi: 10.1007/978-3-030-58595-2_40.

[53] M. Lee, C. Park, S. Cho, and S. Lee, "SPSN: Superpixel prototype sampling network for RGB-D salient object detection," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 630–647.

[54] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon, "Dynamic graph CNN for learning on point clouds," *ACM Trans. Graph.*, vol. 38, no. 5, pp. 1–12, Oct. 2019.

[55] A. Islam, C.-F. R. Chen, R. Panda, L. Karlinsky, R. Feris, and R. J. Radke, "Dynamic distillation network for cross-domain few-shot recognition with unlabeled data," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 3584–3595.

[56] H. Zheng, R. Wang, J. Liu, and A. Kanezaki, "Cross-level distillation and feature denoising for cross-domain few-shot classification," 2023, *arXiv:2311.02392*.

[57] B. Zhao, Q. Cui, R. Song, Y. Qiu, and J. Liang, "Decoupled knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 11943–11952.

[58] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep., SAL-2016-05-27, 2016, vol. 5, p. 5.

[59] L. Liang, S. Zhang, and J. Li, "Multiscale DenseNet meets with bi-RNN for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5401–5415, 2022.

[60] W. Liu et al., "Self-supervised feature learning based on spectral masking for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407715.



Shou Feng (Member, IEEE) received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2019.

He is currently an Associate Professor with Harbin Engineering University, Harbin. His main research interests include remote sensing image processing, data mining, and machine learning.



Hongzhe Zhang is currently pursuing the master's degree with Harbin Engineering University, Harbin, China.

His main research interests include hyperspectral image processing.



Bobo Xi (Member, IEEE) received the B.E. degree in information engineering and the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2017 and 2022, respectively.

He is currently a Lecturer with the State Key Laboratory of Integrated Services Networks, School of Telecommunications, Xidian University. He has published over 15 articles in refereed journals, including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral image processing, machine learning, and deep learning.



Chunhui Zhao received the B.S. and M.S. degrees from Harbin Engineering University, Harbin, China, in 1986 and 1989, respectively, and the Ph.D. degree from the Department of Automatic Measure and Control, Harbin Institute of Technology, Harbin, in 1998.

He was a Post-Doctoral Research Fellow with the College of Underwater Acoustical Engineering, Harbin Engineering University. He is currently a Professor and a Ph.D. Supervisor with the College of Information and Communication Engineering, Harbin Engineering University. His research interests include digital signal and image processing, mathematical morphology, and hyperspectral remote sensing image processing.

Prof. Zhao is a Senior Member of the Chinese Electronics Academy.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is currently a Professor. He is also the Director of the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center. His research interests include image and video processing, hyperspectral image (HSI) processing, and high-performance computing.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He was a Visiting Scholar with Stanford University, Stanford, CA, USA; KTH, Stockholm, Sweden; and NUS, Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He also holds the AXA Chair in remote sensing. He is also an Adjunct Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008. From 2012 to 2017, he was a member of the Institut Universitaire de France. From 2007 to 2010, he was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter that received the 2010 IEEE GRS-S Chapter Excellence Award. He received multiple outstanding paper awards. He was the General Chair of the First IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). From 2009 to 2011, he was the Chair and the Co-Chair of the GRS Data Fusion Technical Committee from 2005 to 2008. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia, from 2017 to 2019. He is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the PROCEEDINGS OF THE IEEE. From 2011 to 2015, he was the Editor-in-Chief of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. He was an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING. In 2014, he served as the Guest Editor for *IEEE Signal Processing Magazine*. He was a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:17:31 UTC from IEEE Xplore. Restrictions apply.